

Mohak Sharma

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RESEARCH INTERESTS

Neural interfaces for embodied agents: decoding human intent from non-invasive physiological signal (EEG, gaze), and the agent-side representations, particularly learned models of an agent's own body, that make sparse intent sufficient to drive high-DOF action. Also: shared autonomy, physics-informed neural fields, neural encoding models.

EDUCATION

Columbia University, M.S. Computer Science

Expected Dec 2026

Coursework: Computational Neuroscience; Computer Vision I/II; Deep Learning; Causal Inference; Mechanistic Interpretability.

Guru Gobind Singh Indraprastha University, B.Tech. Computer Science

2019 to 2023

RESEARCH EXPERIENCE

Laboratory for Intelligent Imaging and Neural Computing (LIINC), Columbia, Graduate Researcher

Spring 2026, Fall 2026 to present

- Designed and ran user studies on intention decoding across multiple interaction modalities: task and stimulus design, IRB protocol support, participant recruitment, instrumentation, and synchronized multi-stream capture of EEG, gaze and behavior.
- Built out the physiological inference pipeline over multimodal neural signals: EEG preprocessing (filtering, artifact rejection, re-referencing, epoching), gaze and pupil feature extraction, temporal alignment across asynchronous streams, and decoders mapping the fused signal onto a latent representation of user intent, with cross-participant generalization as an explicit evaluation criterion.
- Integrated decoded intent with embodied foundation models so that downstream robot action selection and adaptation follow the operator's inferred goal rather than an explicit command. Targeting a complete pipeline demonstration and user study by spring 2027.

Complex Resilient Intelligent Systems Lab (CRIS), Columbia, Research Assistant

2025 to present

- Probed learned representations across 48 transformer layers of a protein structure prediction model over 200K+ AlphaFold DB structures (contrastive representation analysis, UMAP), characterizing how residue-interaction and geometric information is organized with depth; related Predicted Aligned Error structure to interaction interfaces via ChimeraX.

RESEARCH PROJECTS

Neural intent decoding for physics-grounded humanoid control

Independent

- End-to-end pipeline mapping EEG motor imagery to continuous control of a 17-DOF humanoid in MuJoCo: compact convolutional decoder with contrastive self-supervised pretraining, intent decoder emitting joint-space action vectors, PPO/SAC refinement. Trained on PhysioNet EEGMMIDB and Schirrmeister2017; runs end to end under 100 ms, within budget for closed-loop control.

Physics-informed neural fields for body-schema acquisition

Computer Vision II

- Joint-conditioned neural radiance field for a 17-DOF humanoid, regularized by a Hamiltonian energy-conservation term computed from back-projected depth, so reconstructed geometry is mechanically as well as photometrically plausible. Paired with a motor-gated recurrent neural field and spiking-network optical flow; their mismatch flags implausible motion. Tests whether a consistent body schema emerges from visuomotor prediction alone, without body models or labels.

Computational models of neural encoding

Computational Neuroscience

- Time encoding and decoding machines (ASDM, integrate-and-fire) recovering bandlimited signals from spike timings; end-to-end auditory pathway model (cochlear filterbank, hair cell transduction, spiking populations) reconstructing stimuli from population activity; Hodgkin-Huxley and Rinzel synaptic dynamics.

Multimodal intent inference under partial observability

Independent

- Modeled user goals as latent variables in a POMDP spanning screen, VR and console interaction, maintaining a Bayesian belief state over intent and using deep Q-learning for shared autonomy, so the system assists proactively rather than on explicit command.

TECHNICAL SKILLS

Neural signal & modeling: EEG preprocessing (filtering, ICA, re-referencing, epoching), MNE-Python, band power/CSP, spiking neuron models, time encoding machines.

ML: PyTorch, TensorFlow; CNNs, transformers, RNN/LSTM; contrastive and self-supervised learning; RL (PPO, SAC, DDPG); hierarchical Bayesian modeling; causal inference.

Vision, 3D & simulation: Neural radiance fields, volume rendering, multi-view geometry, COLMAP/Open3D, camera calibration; MuJoCo, PyBullet; real-time closed-loop control, POMDPs.

Engineering: Python, TypeScript; FastAPI, React/Next.js, GraphQL; SQL, MongoDB, Redis; AWS, Linux, Git; data and experiment infrastructure.

INDUSTRY EXPERIENCE

Subconscious AI, Founding Engineer, New York

Apr 2026 to present

- Built the inference and insights layer over a hierarchical Bayesian conjoint model (AMCE estimation, kernel-density willingness-to-pay curves, demographic heterogeneity), turning noisy human choice data into decision-grade estimates; migrated the platform from R to React and FastAPI.

Outscal, Growth and Product Engineer, Delhi

Nov 2023 to Jul 2025

- Built an ML job-parsing pipeline processing 2,000+ postings daily by fine-tuning an LLM for structured extraction; designed and ran A/B experiments across the user funnel driving a 15% conversion lift.

TravClan, Frontend Engineer, Delhi

2022 to 2023

- Internal operations platform used by 250+ staff; customer-facing booking portals serving 5,000+ agents.